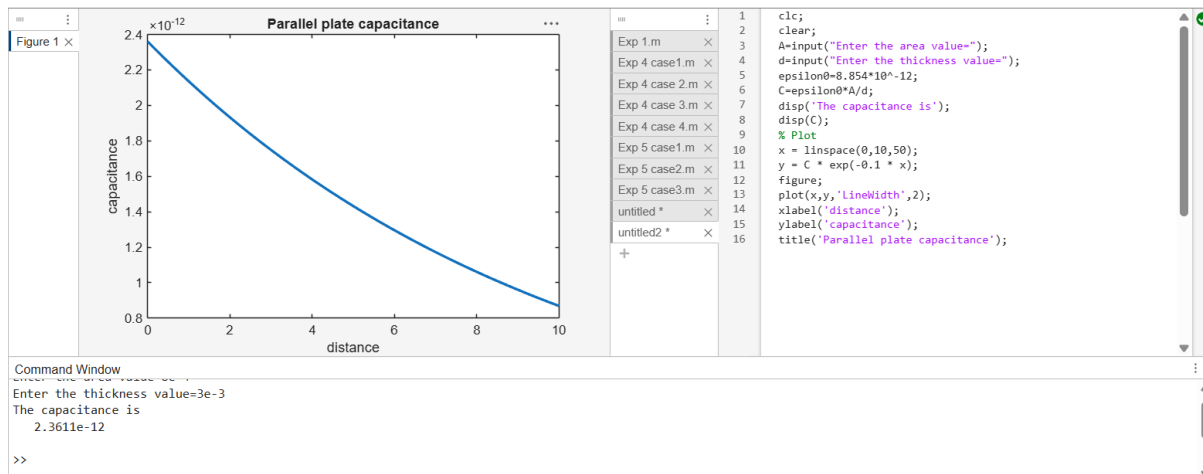


Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constants

Test case (i) Determination and plotting of capacitance of parallel plate capacitor for given area and

dielectric thickness



Test case (ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers

```

1  clc;
2  A=input("Enter the area value=");
3  d1=input("Enter thickness d1=");
4  d2=input("Enter thickness d2=");
5  d3=input("Enter thickness d3=");
6  er1=input("Enter dielectric constant er1=");
7  er2=input("Enter dielectric constant er2=");
8  er3=input("Enter dielectric constant er3=");
9  epsilon0=8.854e-12;
10 C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));
11 disp('The capacitance is');
12 disp(C);
  
```

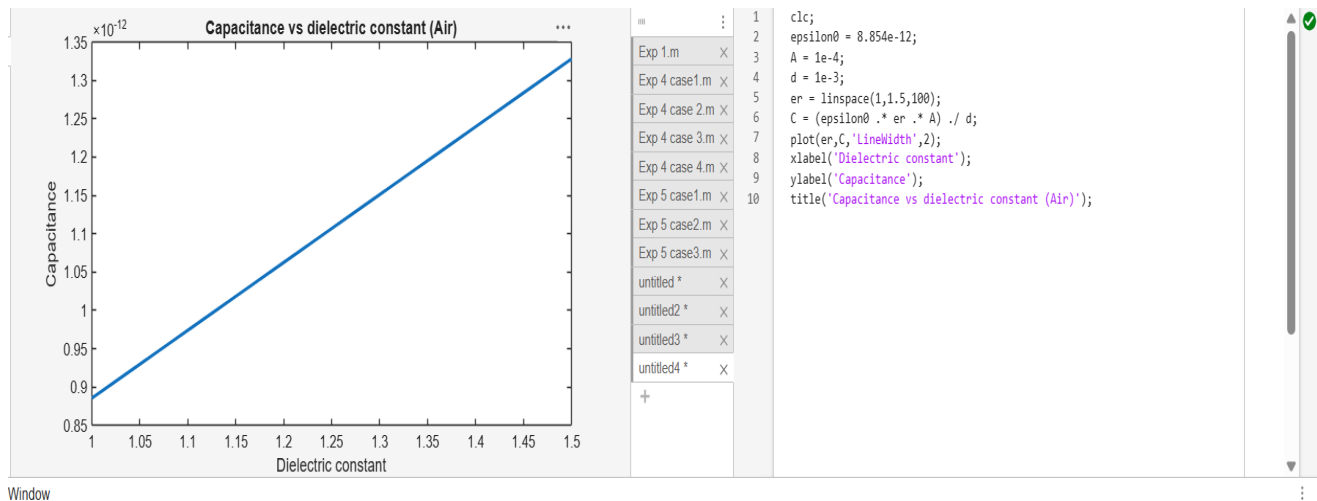
Command Window

```

Enter the area value=8e-4
Enter thickness d1=1e-3
Enter thickness d2=2e-3
Enter thickness d3=3e-3
Enter dielectric constant er1=2.3
Enter dielectric constant er2=3.3
Enter dielectric constant er3=4
The capacitance is
3.9552e-12
  
```

>>

Test case (iii) Determination and plotting of capacitance variation with dielectric constant (Air)



Test case (iv) Determination and plotting of capacitance variation with dielectric constant (Glass)

